

KARRATHA, Australia

WMO: 953070

Lat: 20.7097S

Lon: 116.7742E

Elev: 19

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	53.0	55.1	26.0	20.0	69.4	30.1	24.2	70.2	26.5	68.8	24.7	72.1	7.0	170	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.5	103.0	73.6	100.5	74.2	98.3	74.5	83.4	90.7	82.2	89.9	81.3	89.5	14.5	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.7	165.1	87.5	80.3	157.4	86.7	79.2	151.6	86.4	47.4	90.1	46.1	89.9	45.2	89.7	96.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.2	24.7	22.3	49.0	109.4	2.1	2.5	47.5	111.2	46.3	112.6	45.1	114.0	43.5	115.8 (4)		
(5)			41.5	85.3	1.9	3.1	40.1	87.5	39.0	89.3	38.0	91.0	36.6	93.2 (5)		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.5	87.7	87.5	87.4	83.2	75.2	69.2	68.1	70.1	74.6	81.1	83.8	86.5
	DBStd	8.32	3.77	3.74	3.81	3.54	3.99	4.11	3.63	3.31	3.71	4.87	4.33	4.15
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	24	0	0	0	0	0	9	12	3	0	0	0	0
	CDD50	10760	1168	1051	1159	996	781	577	560	623	737	964	1013	1131
	CDD65	5308	703	631	694	546	316	136	107	160	287	499	563	666
	CDH74	64653	9728	8825	9796	6831	2752	793	687	1321	2691	5674	6775	8780
	CDH80	32311	5367	4886	5538	3392	923	111	87	263	978	2755	3367	4644
Wind	WSAvg	12.0	14.3	12.9	11.8	9.8	9.9	10.1	9.9	10.4	11.8	13.5	14.9	14.8
Precipitation	PrecAvg	9.9	1.0	2.4	1.8	0.8	1.1	1.3	0.6	0.2	0.0	0.0	0.0	0.4
	PrecMax	19.9	4.9	8.6	11.3	4.9	5.9	8.0	3.4	1.3	0.4	0.5	0.3	3.1
	PrecMin	1.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.7	1.4	2.4	2.6	1.3	1.5	2.0	0.8	0.4	0.1	0.1	0.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.9	105.8	104.6	100.6	93.1	86.3	85.8	88.5	95.0	102.5	104.1	105.0
		MCWB	73.7	75.0	73.6	71.7	67.6	67.2	63.4	64.0	66.1	68.7	69.5	71.5
	2%	DB	101.0	100.6	101.1	97.1	90.0	83.1	82.3	85.2	91.2	98.4	99.7	100.8
		MCWB	76.6	77.3	75.0	71.0	67.9	65.3	62.5	63.2	65.2	68.5	71.0	74.0
	5%	DB	97.7	97.7	98.4	94.7	87.4	80.7	80.0	82.9	88.4	95.2	96.5	97.7
		MCWB	78.1	77.5	75.5	71.2	67.7	64.4	62.1	62.6	65.1	69.3	71.6	74.6
	10%	DB	94.9	94.9	95.9	92.5	85.2	78.6	77.9	80.8	85.7	92.1	93.5	94.8
		MCWB	79.0	78.4	75.9	71.5	67.5	63.9	61.8	62.1	64.6	69.2	71.6	75.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	85.7	84.4	84.0	81.5	78.2	75.8	72.5	71.1	73.5	78.9	80.2	82.3
		MCDB	91.9	92.1	91.2	88.0	82.2	78.2	75.2	78.3	81.6	86.5	88.4	90.9
	2%	WB	83.5	82.8	82.5	79.4	76.0	72.4	69.3	69.4	71.4	76.1	78.2	80.9
		MCDB	90.6	90.5	90.1	86.9	81.0	76.5	75.0	77.4	81.1	84.8	87.2	89.8
	5%	WB	82.2	81.8	81.4	78.2	73.9	70.3	67.3	67.7	69.8	74.5	76.8	79.8
		MCDB	89.9	89.8	89.3	86.1	80.5	75.3	74.5	76.3	80.0	84.9	86.9	88.8
	10%	WB	81.1	81.0	80.5	76.8	71.9	68.3	65.3	66.0	68.5	73.1	75.6	78.9
		MCDB	89.4	89.3	88.7	85.4	79.7	74.7	73.4	75.4	79.1	84.4	86.2	87.9
Mean Daily Temperature Range	5% DB	MDBR	13.4	13.5	15.5	18.2	18.6	18.0	20.1	22.3	22.1	20.5	18.0	15.0
		MCDBR	18.0	18.0	19.8	21.5	21.5	20.1	22.6	24.6	25.8	24.1	22.6	19.4
		MCWBR	7.4	7.7	8.4	9.8	10.3	10.0	11.2	11.8	11.5	10.0	9.2	8.0
	5% WB	MCDBR	13.8	14.5	15.1	16.5	16.4	14.0	17.8	21.0	22.5	19.8	17.6	14.4
		MCWBR	6.6	6.9	6.9	8.2	9.5	8.2	10.7	11.1	10.5	9.6	8.5	6.6
Clear-Sky Solar Irradiance	taub	0.402	0.397	0.378	0.353	0.335	0.322	0.317	0.314	0.333	0.357	0.359	0.378	
	taud	2.506	2.533	2.575	2.600	2.586	2.609	2.596	2.565	2.512	2.472	2.505	2.524	
	Ebn at Noon	299	297	295	290	284	282	287	298	304	305	310	307	
	Edh at Noon	36	35	32	29	27	26	27	30	34	37	36	36	
All-Sky Solar Radiation	RadAvg	2368	2265	2023	1763	1475	1336	1451	1793	2119	2394	2552	2547	
	RadStd	143	155	144	94	113	95	93	50	80	54	61	102	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (8 neighbors)	+0.83	N/A	N/A	N/A	+0.86	+1.01	N/A	N/A	N/A	+284

Nomenclature: See separate page